

Use MQTT Protocol In Smartphones And Raspberry Pi

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The article demonstrates the use of Message Queuing Telemetry Transport (MQTT) protocol in smartphones and Raspberry Pi.

MQTT, a machine-to-machine Internet of Things (IoT) protocol, is useful for connecting with a remote client where a small code footprint or very low Internet bandwidth is needed. It has been used in sensors communicating to a broker/server using satellite link, and in a range of home

automation and small devices. It is ideal for mobile applications because of its low power usage, small size, minimised data packets and proper distribution of information to one or many receivers.

MQTT protocol was first designed in 1999. With the growth of the IoT and the need to communicate between low-powered devices, it has become quite popular in recent times.

How MQTT works

MQTT protocol uses a publish/subscribe architecture unlike HTTP, which uses request/response architecture. MQTT protocol is event-driven and enables messages to be pushed

to clients. The heart of MQTT protocol is MQTT broker (server), which is responsible for dispatching messages between senders and receivers (clients).

Every client that publishes a message to the broker includes a topic/information into that message. Each client that wants to receive a message needs to subscribe to a particular topic. The broker delivers the message with the matching topic to a particular client. Clients do not need to know each other as they only communicate using the topic over MQTT broker. The block diagram of the MQTT communication protocol is shown in Fig. 1.

Understanding MQTT using a practical example

There is a free server (<http://test.mosquitto.org/>) where you can test your first MQTT message. Default port for MQTT is 1883; since it is not encrypted (not secure), it is not advisable for commercial usage.

You need two Android phones or two PCs with an Internet connection on both. For now, let us use two Android phones (phone1 and phone2) and follow the steps given below.

1. Download MQTT Client (Web-neurons) application on both phones from the link <https://play.google.com/store/apps/details?id=com.deepeshc.mqttrec>

2. Open the application and enter the details under Settings, as shown in Fig. 2.

In first text box, enter io.kotaielectronics.com URL of MQTT server (broker). In the second line/text box, enter port as 1883. Leave the username and password spaces blank. Press Save and then Connect. Enter the same details on the second phone.

3. You will see green colour on the top-left corner, which indicates that

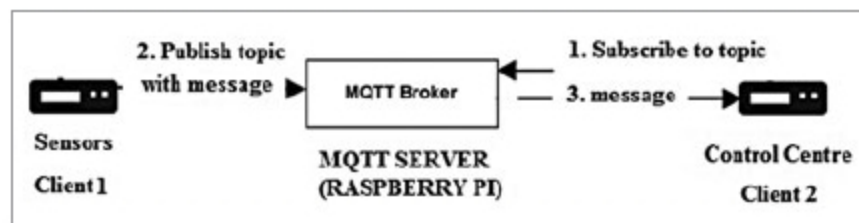


Fig. 1: Block diagram of MQTT communication protocol

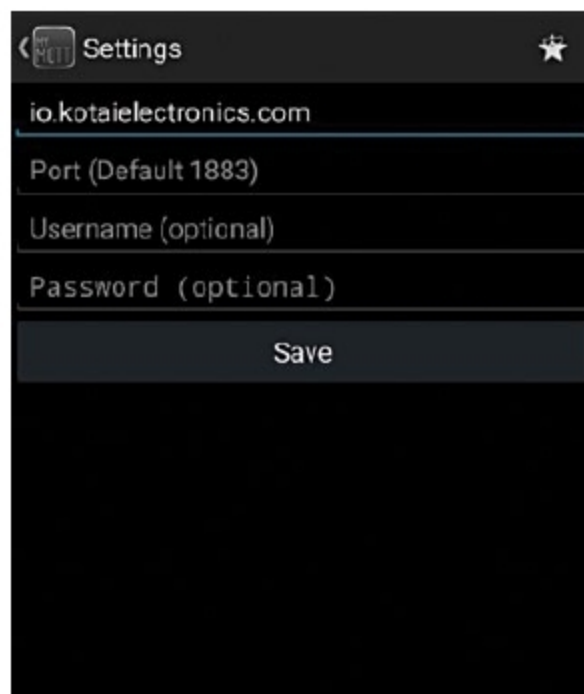


Fig. 2: Settings in Android phone1

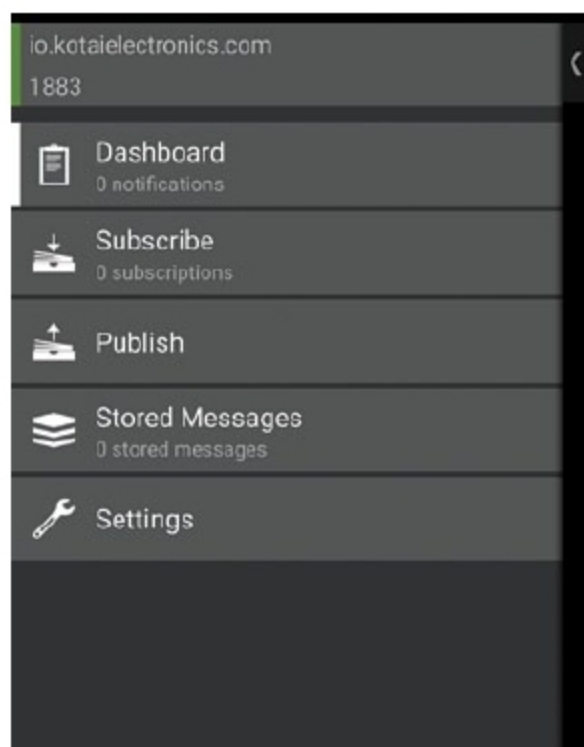


Fig. 3: MQTT connected

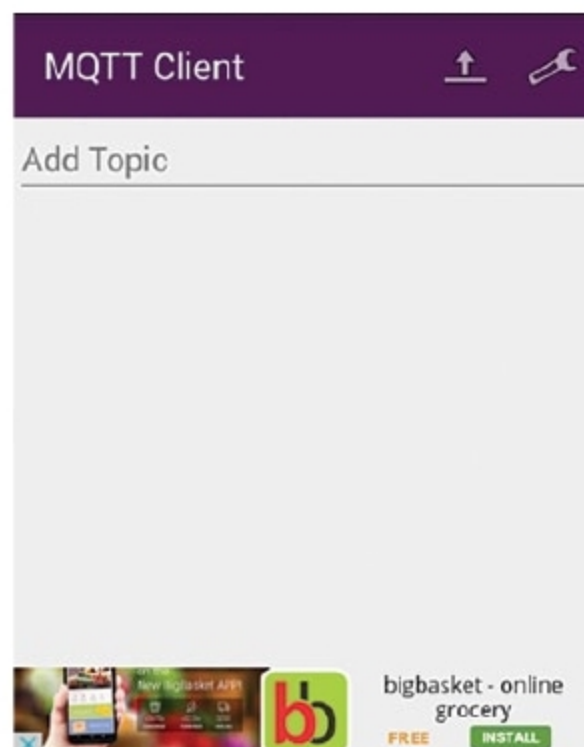


Fig. 4: Add topic in phone1